Power BI: Theoretical Questions and Answers

1. Power BI’da visual nima?

Visual — ma’lumotlarni grafik, jadval, diagramma yoki boshqa ko‘rinishda aks ettiruvchi element.

2. Dashboard va report o‘rtasidagi farq nimada?

Dashboard — bir nechta muhim ma’lumotlarni bitta sahifada ko‘rsatadi. Report esa bir yoki bir nechta sahifadan iborat batafsil tahlildir.

3. Qaysi holatda Table visual ishlatiladi?

Ma’lumotlarni satr va ustunlar ko‘rinishida batafsil ko‘rsatish kerak bo‘lganda ishlatiladi.

4. Qaysi holatda Bar chart ishlatiladi?

Kategoriyalar yoki qiymatlarni bir-biri bilan taqqoslash kerak bo‘lganda ishlatiladi.

5. Slicer nima va nima uchun ishlatiladi?

Slicer — ma’lumotlarni tanlangan qiymat bo‘yicha filtrlash uchun ishlatiladigan interaktiv filter.

6. Visual ichida filter qilish qanday ishlaydi?

Filter tanlanganda visual faqat belgilangan shartga mos ma’lumotlarni ko‘rsatadi.

7. Measure vizuallarda qanday rol o‘ynaydi?

Measure hisoblangan natijalarni, masalan Total Sales yoki Average Sales qiymatlarini visualda ko‘rsatadi.

8. Nima uchun DAX visual bilan birga ishlatiladi?

DAX yordamida Measure va hisob-kitoblar yaratiladi, natijalar esa visual orqali ko‘rsatiladi.

9. Aggregation (SUM, AVERAGE) visualda qanday ishlaydi?

SUM qiymatlarni yig‘adi, AVERAGE esa ularning o‘rtacha qiymatini hisoblaydi.

10. Cross-filtering nima?

Cross-filtering — bitta visualdagi ma’lumot tanlanganda boshqa vizuallarning ham shu tanlov asosida filtrlanishidir.

2. DAX Tasks

Calculated Column

1. FullLocation ustuni yarating:

City & " - " & RoadName

Quyidagi kod orqali “FullLocation” nomli yangi ustun yaratildi.

```
FullLocation = traffic_congestion_data[City] & " - " & traffic_congestion_data[RoadName]
```

FullLocation
Tokyo - Shibuya Crossing
Osaka - Namba Street
London - Oxford Street
New York - 5th Avenue
Seoul - Gangnam Road
Berlin - Unter den Linden
Paris - Champs Elysees
Dubai - Sheikh Zayed Road
Singapore - Orchard Road
Istanbul - Taksim Square
Tokyo - Shinjuku Avenue
Osaka - Umeda Road
London - Baker Street
New York - Broadway
Seoul - Itaewon Road
Berlin - Friedrichstrasse
Paris - Rue de Rivoli
Dubai - Jumeirah Road
Singapore - Marina Bay Road
Istanbul - Besiktas Road
Tokyo - Ginza Street
Osaka - Tennoji Road

2. Year ustuni yarating:

YEAR(Date)

Year ustunini yaratish kodi: Year = YEAR(traffic_congestion_data[Date])



Year
2024
2024
2024
2024
2024
2024
2024
2024
2024
2024
2024

3. Month ustuni yarating:

MONTH(Date)

Year ustunini yaratish kodi:

Month = MONTH(traffic_congestion_data[Date])



Month
1
1
1
1
1
1
1
1
1
1
1
1

4. VehicleType ustunini katta harfga o‘tkazing

```
VehicleType_Upper = UPPER(traffic_congestion_data[VehicleType])
```

VehicleType_Upper
CAR
BUS
CAR
TAXI
CAR
BUS
CAR
TRUCK
CAR
BUS

5. RoadName uzunligini hisoblang

```
RoadName_Length = LEN(traffic_congestion_data[RoadName])
```

RoadName_Length
16
12
13
10
12
16
14
17
12
13
15

6. Traffic category ustuni yarating:

agar CongestionLevel = "High" bo‘lsa → "High Traffic", aks holda "Normal"

```
Traffic_Category =
IF(
    traffic_congestion_data[CongestionLevel] = "High",
    "High Traffic",
    "Normal"
)
```

Traffic_Category
High Traffic
High Traffic
High Traffic
High Traffic
Normal
Normal
High Traffic
Normal
Normal
High Traffic
High Traffic

Measure

7. Total TrafficVolume measure yarating

```
Total TrafficVolume = SUM(traffic_congestion_data[TrafficVolume])
```



8. Average Speed measure yarating

```
Average Speed = AVERAGE(traffic_congestion_data[AverageSpeed])
```



9. Total SignalDelay measure yarating

```
Total SignalDelay = SUM(traffic_congestion_data[SignalDelay])
```



10. Record count measure yarating

```
Record count = COUNTROWS(traffic_congestion_data)
```



11. Max CongestionIndex measure yarating

```
Max CongestionIndex = MAX(traffic_congestion_data[CongestionIndex])
```



12. Min CongestionIndex measure yarating

```
Min CongestionIndex = MIN(traffic_congestion_data[CongestionIndex])
```



3. Visual Tasks

1. Table visual yarating:

o City

o RoadName

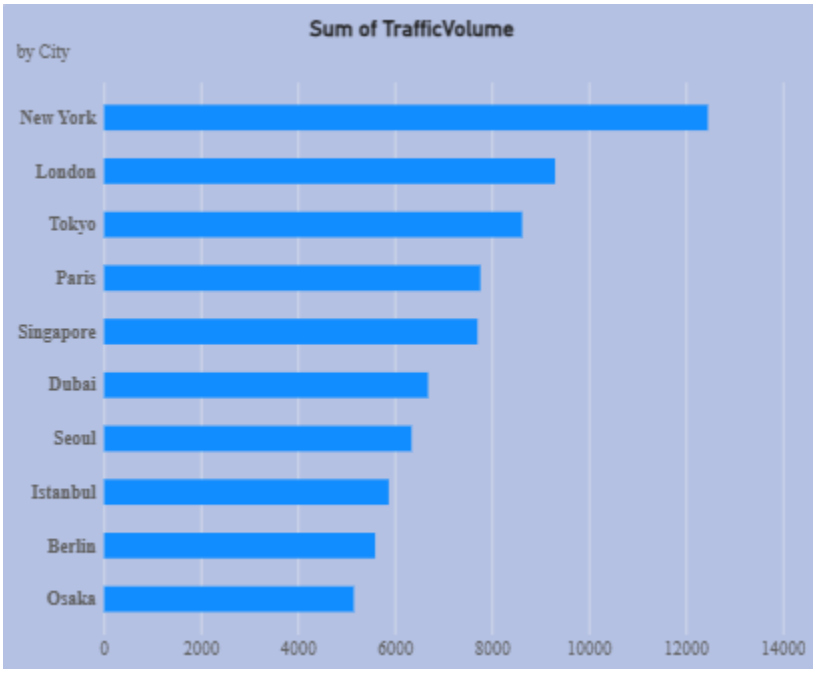
o TrafficVolume

City	RoadName	Sum of TrafficVolume
New York	5th Avenue	2000
Tokyo	Akihabara Street	1400
Dubai	Al Barsha Road	1130
Dubai	Al Khail Road	1150
Dubai	Al Wasl Road	1120
Berlin	Alexanderplatz	920
Paris	Avenue Montaigne	1220
London	Baker Street	1450
Istanbul	Besiktas Road	980
Istanbul	Bosphorus Road	1000
Paris	Boulevard Saint Germain	1350
New York	Broadway	2100
Singapore	Bugis Street	1300
London	Camden Street	1620
Paris	Champs Elysees	1400
Berlin	Charlottenburg Road	930
Total		**75460**

2. Bar chart yarating:

o Axis: City

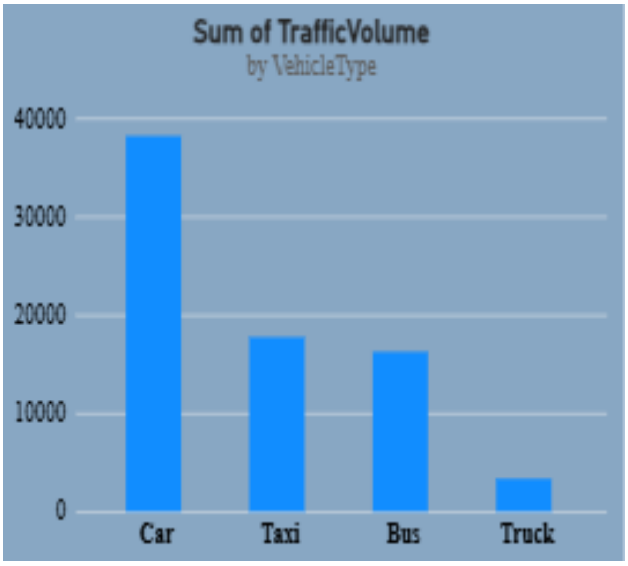
o Values: Total TrafficVolume



3. Column chart yarating:

o Axis: VehicleType

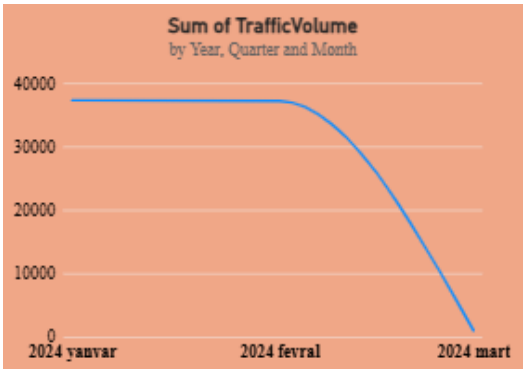
o Values: Total TrafficVolume



4. Line chart yarating:

o Axis: Date

o Values: Total TrafficVolume



5. Card visual yarating:

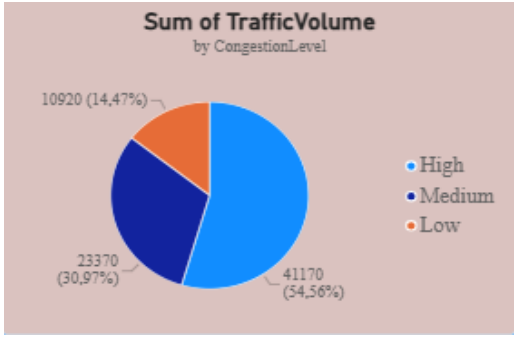
o Total TrafficVolume



6. Pie chart yarating:

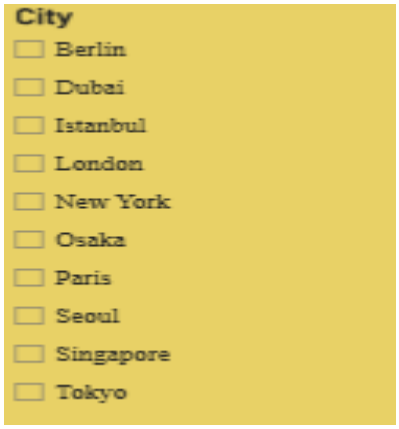
o Legend: CongestionLevel

o Values: TrafficVolume



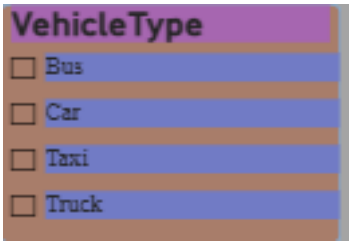
7. Slicer qo‘shing:

o City

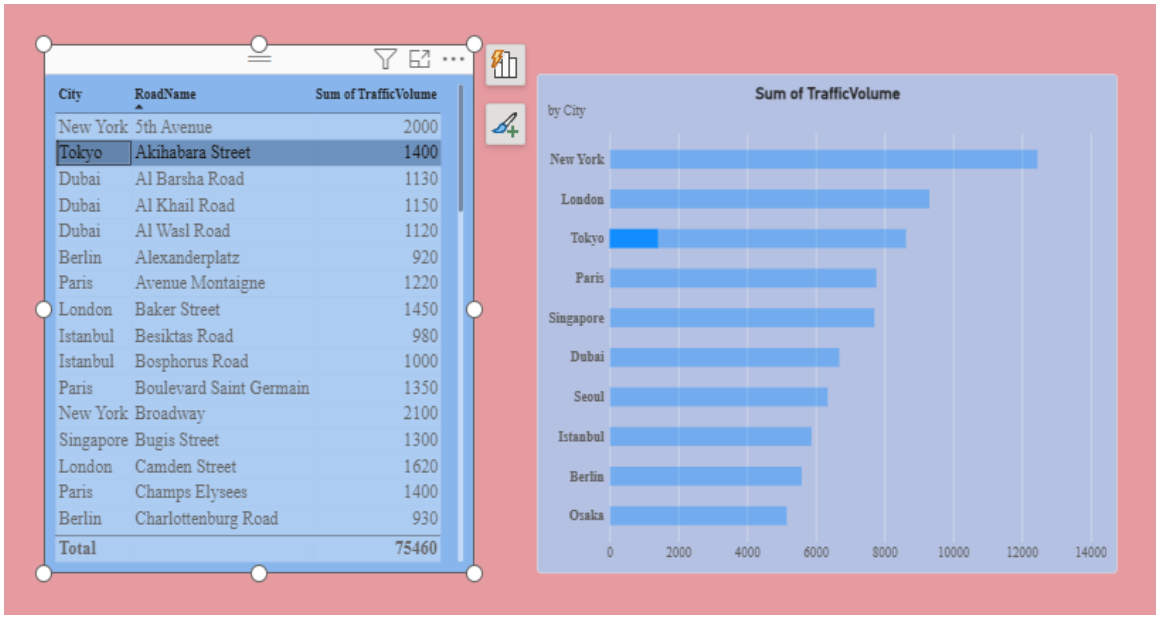


8. Slicer qo‘shing:

o VehicleType



9. Visual’lar o‘zaro qanday bog‘lanishini kuzating (cross-filtering)



10. Vizualni formatlang:

- Title qo‘shing
- Ranglarni o‘zgartiring
- Data labels yoqing

Har bir vizualga ranglar, title va data labelslar mos ravishda shakllantirilgan.